



DSA Course With Placement Prep

For Freshers & Students

Final Year Project Assistance

Crack interviews in
product-based MNC's



www.learnbay.co

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Program Summary

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Training Mode

100% Live Online Session with Instructors from top companies

2

Program Faculty

Renowned **industry experts** from top global MNCs

3

Program Duration

3-4 Months Program Weekday and Weekend Batch

4

150+ Practice Problems

Practice **interview problems** of top tech companies.

Build your foundation for a
Bright Tech Future!



Placement Prep For Top MNCs:

Improve Your Problem-Solving Skills with DSA & Aim for Top Tech Companies



Live Interactive Sessions:

Hands-on coding sessions and Companywise Interview Preparation.



Instructors from MAANG:

Learn from professionals with extensive teaching experience.

About Course

This program is designed for **freshers and beginners** aiming to build a strong foundation in **software development** and prepare for roles at top tech companies. Through a **hands-on, project-based approach**, the course focuses on real-world problem-solving.

The curriculum focuses on **foundational concepts in Data Structures and Algorithms (DSA)**, taught under the guidance of experienced mentors. By the end of the program, you'll **gain the essential skills and confidence** to solve technical problems effectively and take the **first step toward a successful career** in software engineering.

Our Commitment

"We are dedicated to providing accessible, practical, and industry-aligned education that empowers **India's future tech leaders** to excel and grow."

We offer flexible learning options, including **100% online classes** paired with **personalized one-on-one doubt-clearing sessions** to ensure a seamless learning experience.

To further support your journey, we provide comprehensive career guidance with services such as resume building, interview preparation, and skill enhancement—helping you confidently take the first step into a software development role at top product-based companies.



100%

Placement Assistance



350+

Hiring Partners



7+

Centers Across India



40k+

Learners Upskilled

Program Eligibility

Designed for
students and
fresh
graduates

Beginner-Level

- This program provides a strong foundation in Data Structures and Algorithms.
- Ideal for those preparing for their first **technical interviews**.



Final year students/ Fresh grads



Freshers (Beginners)

Program Outcome: What's in it for you?



Master Core DSA Concepts

Develop a strong foundation in Data Structures and Algorithms through hands-on learning, preparing you to solve technical challenges confidently.



Practical Problem-Solving Skills

Apply your knowledge to real-world scenarios with structured exercises and coding challenges designed to build expertise in problem-solving and algorithmic thinking.



Build a Strong Career Foundation

Develop the skills necessary for entry-level software engineering roles with personalized guidance from mentors.



Why choose Learnbay?

A unique program
for Freshers!

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1. Placement Preparation for Freshers

Get job-ready with resume building, technical mock interviews, HR rounds, and project-based interview prep.

Learn to optimize your **Github profile**, tackle real coding challenges, and confidently present your skills to land your first software engineering role. 🚀

Placement Assistance with:

- ✓ 5 Technical Mock Interview
- ✓ 2 HR Mock Rounds

2. Final Year Project Assistance

Get expert guidance for your **Final Year Project!** From idea selection to implementation, we help freshers build industry-ready projects with hands-on support.



**Live Projects
from industry**



**Dedicated
Project Mentor**

3. Faculty and Mentors From MAANG

Learn from seasoned mentors with industry experience who simplify complex concepts and provide personalized support throughout your learning journey.

Trainers from:



Microsoft



4. Get certified and accelerate your career growth

Learnbay Course Completion Certificate



Practical Validation:

The certification reflects not only theoretical knowledge but also the **practical implementation of concepts** through live projects and case studies.

Alumni Spotlight



Shishir Kamal

I'm currently enrolled in the Full Stack course and have completed the DSA and System Design modules. The trainers' teaching method has been enlightening for me as a beginner. Highly recommend Learnbay for upskilling.



Arpit Agarwal

This online software development course was the best I've taken. Great instructor, easy-to-understand explanations, well-structured and effective hands-on exercises. Highly recommended!



Ritam Mukherjee

Great course for software development, with real-world examples and practical exercises. I learned a lot and could apply it in my career. Highly recommend for learners.



Amrita Panjwani

Highly recommend course for software development. Well-organized material, practical exercises to apply and build skills. Suitable for anyone interested in the field.

Alumni Spotlight



Jatin Kumar Khilrani

The software development course was a game-changer. Extensive content, highly skilled trainer, exceptional job support laid the foundation for my career in software development.



Tushar Singhal

Learnbay's software course surpassed my expectations, with clear instruction and helpful instructors. Grateful for the gained knowledge and accessible support.



Gowthaman Jeganathan

I'm currently enrolled in the Full Stack course and have completed the DSA and System Design modules. The trainers' teaching method has been enlightening for me as a beginner. Highly recommend Learnbay for upskilling.



Abhishek Pakhide

I recently finished Learnbay's Basic DSA Course online. The course is well-structured, taught by knowledgeable trainers with a focus on interview preparation and data structure concepts. It benefits both novices and experts.

Alumni Spotlight



Qudsia Tahniyath

Learnbay has helped me a lot to learn data science applications in the e-commerce industry. The live class concept was really helpful in receiving proper DS training. Thanks to all my mentors and the placement team.



Binit Kumar Swalsingh

I've been with this org since Jan 4, 2023, studying Full Stack Software Development. If you have ambition and need a mentor to show you the right path to success with the right knowledge, I highly recommend this org.



Swarup Halder

I have enrolled in Software Development course. I had basic knowledge of DSA and since then it has been a wonderful learning experience. The teacher does an excellent job of explaining the concepts in a way that is easy to understand.



Karanveer Bansal

I enrolled in Learnbay's full stack program. I already completed basic DSA and the live sessions were clear, with good teaching pace. They offer an easy and simple EMI option for course fees.

Program Fee

Live online classes

Benefits:

- ✓ All classes are covered in live interactive mode
- ✓ Schedule doubt clearing session with industry expert

Program Fee

₹ 45,000 + 18% GST

Pay in easy EMIs starting as low as

₹ 4,425/ month

Book Your Demo Class Now!

Experience our engaging teaching methods, connect with seasoned industry experts, and get **personalized guidance** to shape your career path.

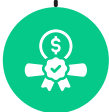
Why attend the Demo Class?



Interactive Learning: Get a first-hand look at how we simplify complex topics like DSA.



Mentor Guidance: Receive insights from industry professionals about how to prepare for tech interviews



Scholarship Opportunity: Take the assessment test and qualify for up to 30% scholarship based on your performance.

[Book a demo now](#)



Program Curriculum

A unique program
for Freshers!

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Topic 1 Programming Fundamentals

Duration: 2-3 weeks

Outcome of this term: This module helps engineers brush up on fundamental programming skills, ensuring a strong foundation before diving into advanced topics. If you're already comfortable with the basics and actively coding, **you can skip this** and jump straight into the **DSA & Problem Solving**.

Topic 1: Introduction to Programming

- Basics of programming and Java installation (**JDK, IDE like IntelliJ, Eclipse**)
- Writing and running your first Java program (e.g., Hello World)

Topic 2: Variables, Data Types, and Operators

- **Variables:** Declaration, Initialization, Scope in Java
- **Data types:** int, float, double, char, boolean
- **Operators:** Arithmetic, Relational, Logical, Increment/Decrement
- Input and output in Java: **Scanner class and System.out.println**

Topic 3: Control Flow

- Conditional statements: if, else if, else
- **Switch-case** in Java
- **Loops:** for, while, do-while
- Break and continue

Topic 4: Functions (Methods)

- Defining and calling methods in Java
- Method parameters, return types, and scope
- Method overloading
- **Recursion in Java**

Topic 5: Arrays and Strings

- **Arrays:** Declaration, initialization, and traversal
- **Multidimensional arrays**
- **Strings:** String class methods, concatenation, comparison, and manipulation

Topic 6: Object-Oriented Programming (OOP)

- Classes and Objects
- **Constructors** and destructors
- Encapsulation, **Inheritance**, Polymorphism, Abstraction
- Access modifiers: private, public, protected
- Static and non-static members
- **OOP practice problems (e.g., real-world object modeling)**

Note:

"Mastering the fundamentals is key to understanding complex concepts later. Treat this as the bedrock of your coding journey."

To Do:

Create and debug a simple program that performs basic arithmetic operations using user input.

Topic 2 Data Structures & Algorithms

Duration: 2 Months

Outcome of this term: This module helps you master **DSA and problem solving skills** by solving real problems like those in tech interviews. You'll learn to work with *arrays, linked lists, trees, graphs*, and more, focusing on building efficient solutions. Gain confidence to tackle challenging problems and **succeed in technical interviews**.

Section 1: Foundations of Data Structures and Algorithms

Time and Space Complexity

- Understanding Performance: Importance of complexity analysis in interviews.
- **Big O Notation:** Common complexities and how to analyze them.
- **Space Complexity:** Optimizing memory usage in data structure choices.

Arrays

- **Array Basics:** Handling and manipulating arrays.
- Operations: Insertion, deletion, searching, and updating,
- Two-Dimensional Arrays: Matrix manipulation.
- Maximum element in an array, **Reverse an array**, Minimum element in an array.
- Sorted array check, Count occurrences.
- Rotate an array, Merge sorted arrays, Intersection of arrays, Move zeroes, Pairs with given sum
- **Longest increasing subsequence**, Maximum subarray sum.
- Longest common subsequence, Minimum path sum, Maximum product subarray

Linked Lists & Interview Problems

- **Singly Linked List:** Creation, traversal, insertion, and deletion.
- **Doubly and Circular Linked Lists:** Differences and use cases; related problems.
- **Sample Practice Problems:**
- Middle element of a linked list, **Detect a loop in a linked list**, Length of a linked list, Check if a linked list is a palindrome, Merge two linked lists

Stacks and Queues

- Understanding Concepts: **LIFO** and **FIFO** principles—common in problem statements.
- Operations: Implementing stacks and queues efficiently.
- **Priority Queues and Dequeues:** Applications in interview questions.
- **Sample Practice Problems:**
- Check for balanced parentheses, Implement a stack using linked list, Implement a stack using an array,

Section 2: Advanced Data Structures and Algorithms

Recursion and Backtracking

- Mastering Recursion: Key concepts and base cases crucial for problem-solving
- **Divide and Conquer:** Understanding its application in interviews.
- **Backtracking:** N-Queens, Sudoku.

Searching Algorithms

- **Linear vs Binary Search:** Time complexities.
- Advanced Searches: Ternary search and applications.

Assignments:

Solve at least 5 problems from LeetCode or HackerRank focusing on arrays and strings.

Sorting Algorithms

- Simple Sorting Algorithms: **Bubble**, selection, insertion—understanding their performance.
- **Advanced Sorting Algorithms**: Merge sort, quick sort—key algorithms in interviews.

Hashing

- **Hash Tables and Hash Functions**: Importance in optimizing search operations.
- Collision Handling Techniques: Techniques to handle collisions effectively.
- Hash Table Implementations

Mini Challenge:

Reverse a linked list using recursion and iteratively—compare the two approaches.

Mastering Trees for Problem Solving

- **Binary Trees**: Understanding traversal techniques (inorder, preorder, postorder).
- **Binary Search Trees** (BSTs)
- Balanced Trees (AVL, Red-Black): Importance of balancing in interviews.
- **Practice Problems**:
- Implement binary tree traversals, Count the number of nodes in a binary tree.

Understanding and Implementing Tries

- **Understanding Tries:** Structure and its applications in string problems.
- Insertion and Search Operations
- **Practice Problems:**
 - Insert and search in a trie.
 - Find all words with a given prefix, **Count distinct substrings.**

Heap Structures and their Applications

- **Max-Heaps and Min-Heaps:** Key properties & operations.
- Heap Applications
- **Practice Problems:**
 - Build a max heap from an array, Find the maximum element in a heap.

Greedy Algorithms

- **Understanding Greedy Strategy:** When to apply greedy methods in problem-solving.
- **Practice Problems:**
 - Coin change problem (greedy version), Activity selection problem.

Dynamic Programming (DP)

- **Introduction to DP:** Recognizing overlapping subproblems and optimal substructure.
- Top-down vs. Bottom-up:
- **Common DP Problems:** Longest common subsequence, 0/1 knapsack

Graphs

- **Graph Representations:** Adjacency matrix vs. adjacency list—pros and cons.
- Graph Traversal Algorithms: BFS and DFS—understanding their applications in problem-solving.
- **Shortest Path Algorithms:** Dijkstra's, Bellman-Ford; how to apply them in interviews.

Placement Preparation Module for Freshers

Resume Building & Profile Optimization

- Craft an **ATS-friendly** resume tailored for entry-level roles.
- Optimize LinkedIn and GitHub profiles for better visibility.
- Learn how to **highlight projects** and technical skills effectively.

Mock Interviews & HR Rounds

- **5 Technical Mock Interviews** to simulate real coding rounds.
- **2 HR Interviews** to prepare for behavioral and situational questions.
- **Personalized feedback** to improve communication and confidence.

Project-Based Preparation

- Learn to present your projects effectively during interviews.
- Develop structured explanations for problem-solving approaches.
- Get guidance from industry experts for **final year project**.

Outcome:

Get interview-ready with structured practice, real-time feedback, and polished presentation skills! 🚀

Thank you!

For more queries and
information please reach out
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Visit us at

www.learnbay.co

